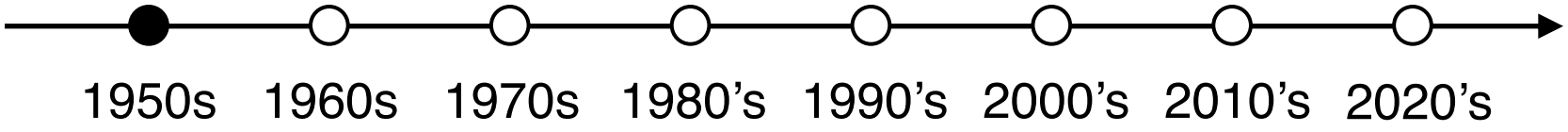
Introduction to Mutimodal AI for Biomedical Applications

[W1-2: Introduction to AI]

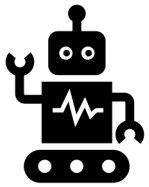
Jun-Hyuk Kim

Department of AI

History of AI



- 1950: Alan Turing's Turing Test
  - To check if machines seem human



"I am human"



"I am human too"



Human
(Evaluator)

Alan Turing
(1912 – 1954)



History of AI

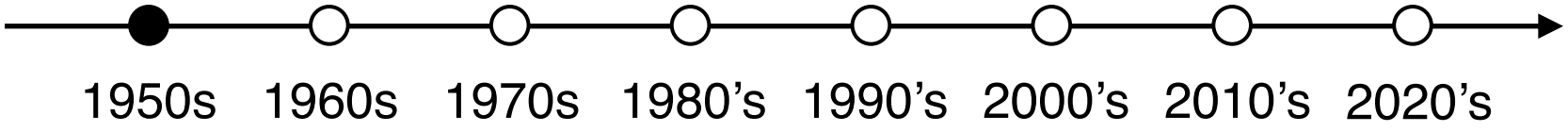


- 1956: Dartmouth Conference
  - 10 researchers attended
  - Proposed the name "Artificial Intelligence"
  - Worked on calculating possibility
    - Not today's advanced AI
    - But led to AI we have now

John McCarthy
(1927 – 2011)



History of AI

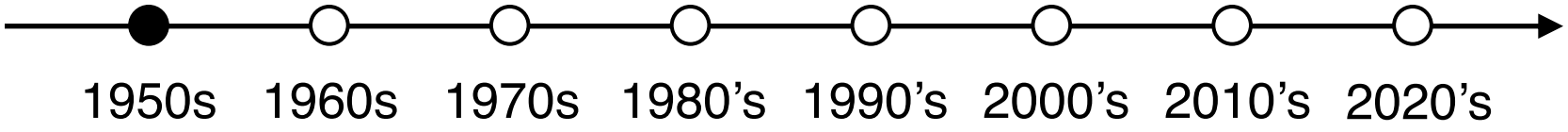


- 1958: Perceptron
  - Early attempt to mimic human brain neurons
  - Designed to recognize simple pattern
  - Machines could improve with practice
    - Inspiring later AI ideas

Frank Rosenblatt
(1928 – 1971)



History of AI

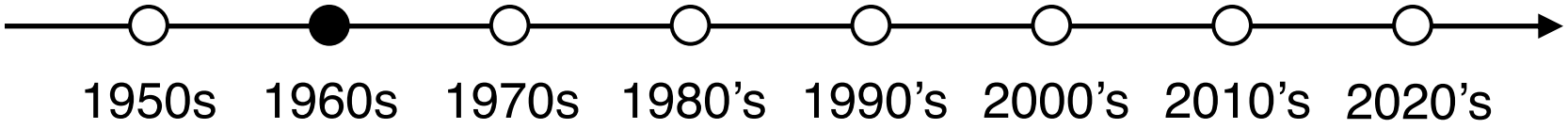


- Before Perceptron
  - Marvin Minsky led rule-based AI (Symbolism)
- Perceptron
  - Machines that learn from data (Connectionism)
- Symbolism < Connectionism

Marvin Minsky
(1927 – 2016)

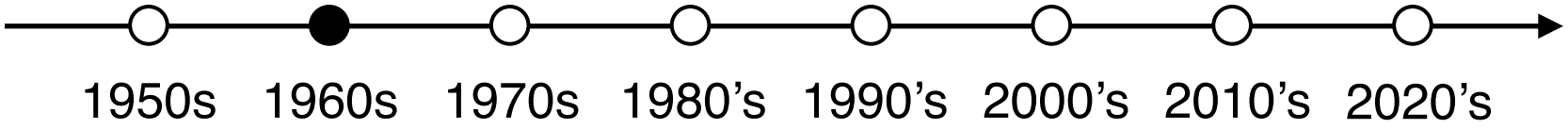


History of AI



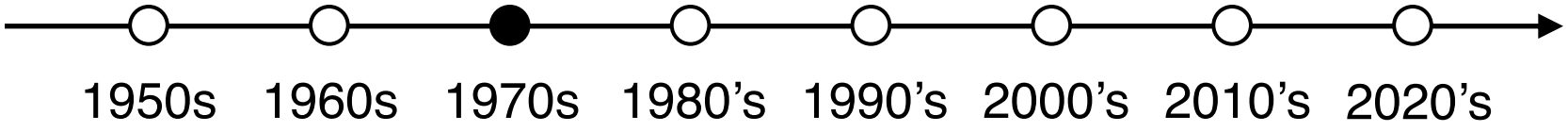
- In 1969, Marvin Minsky showed the Perceptron's limits
- Attention returned to Symbolic AI for a while
- Symbolism > Connectionism

History of AI



- In 1969, Marvin Minsky's Critique
  - Perceptron could solve OR, but not XOR
    - OR: (0, 1), (1, 0), (1, 1) → 1; (0, 0) → 0
    - XOR: (0, 1), (1, 0) → 1; (0, 0), (1, 1) → 0
  - Showed Perceptron couldn't handle complex patterns

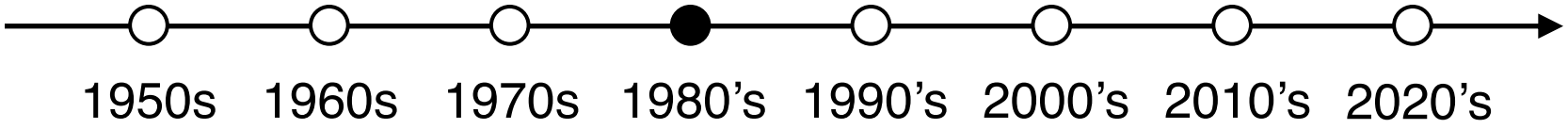
History of AI



- 1970s: The First AI Winter
  - Rule-base AI's limits
  - Learning-based AI's limits
  - Failed to meet expectations
  - Research slowed down

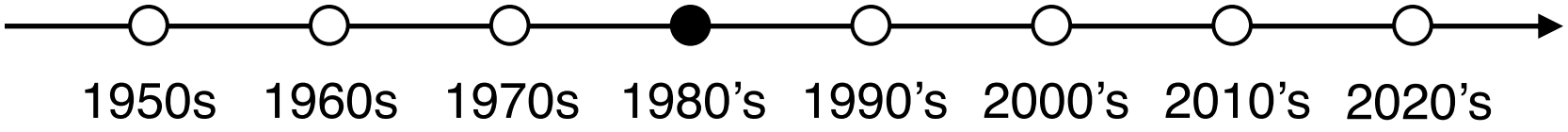


History of AI

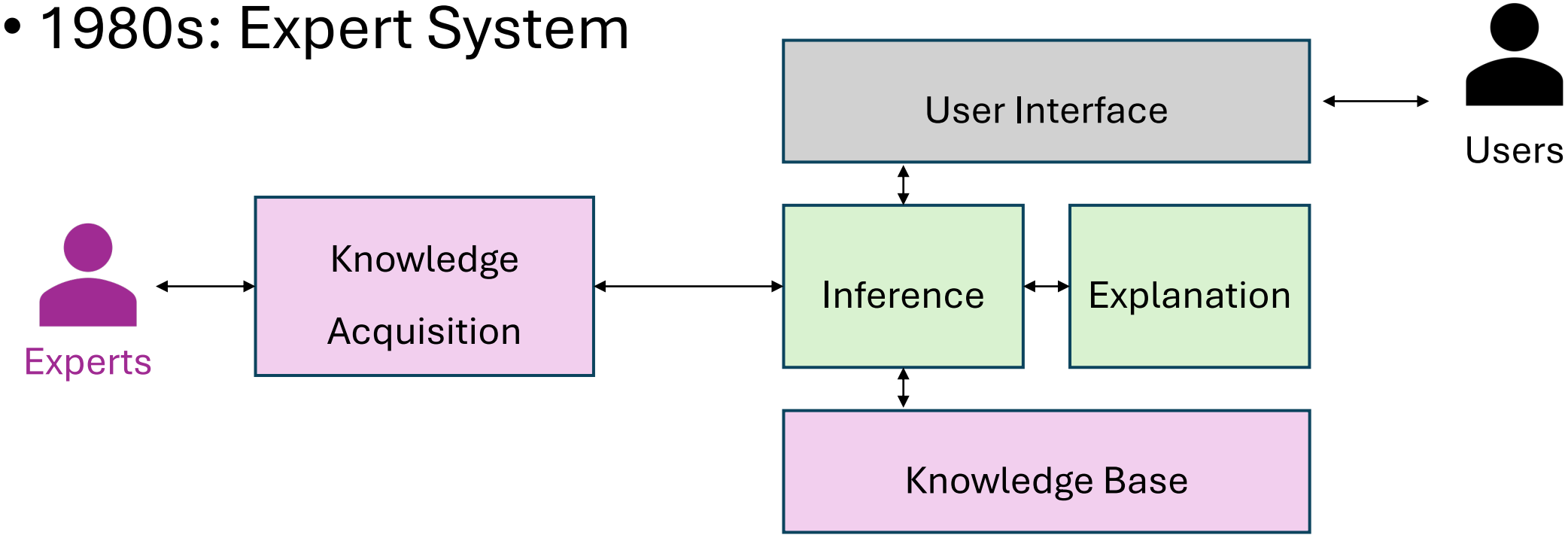


- 1980s: Expert System
  - Rule-based AI that mimic human experts in specific fields
  - Examples: Doctors, lawyers, enginners

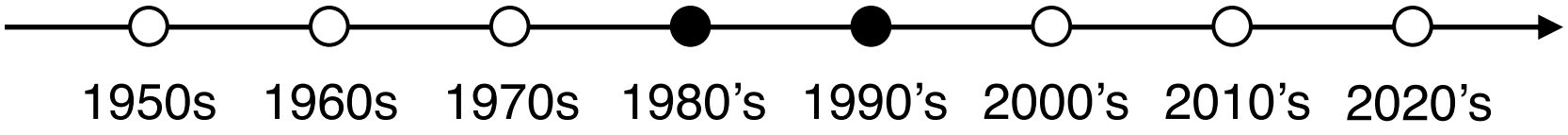
History of AI



- 1980s: Expert System



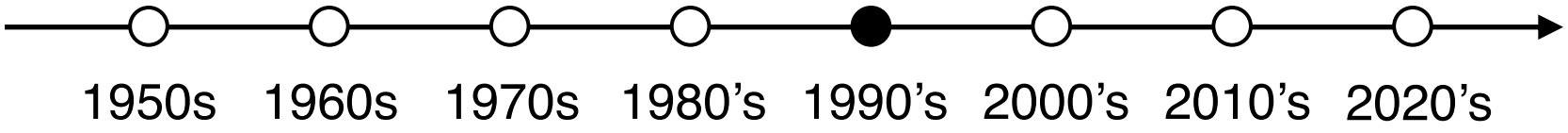
History of AI



- 1987 – 1993: The Second AI Winter
  - Too many rules to manage
  - High maintenance costs
  - Poor adaptability
  - Limited cost-effectiveness

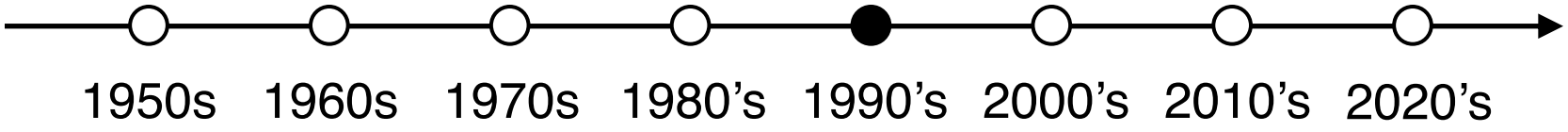


History of AI



- 1990s: Machine Learning
  - Machines learn from data instead of following human's rules
  - Development of various machine learning algorithms
  - Examples: SVM (1995), AdaBoost (1995)

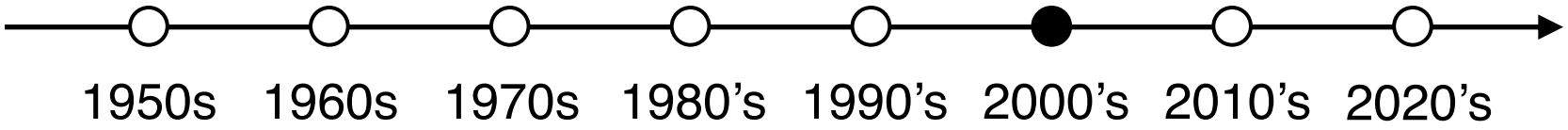
History of AI



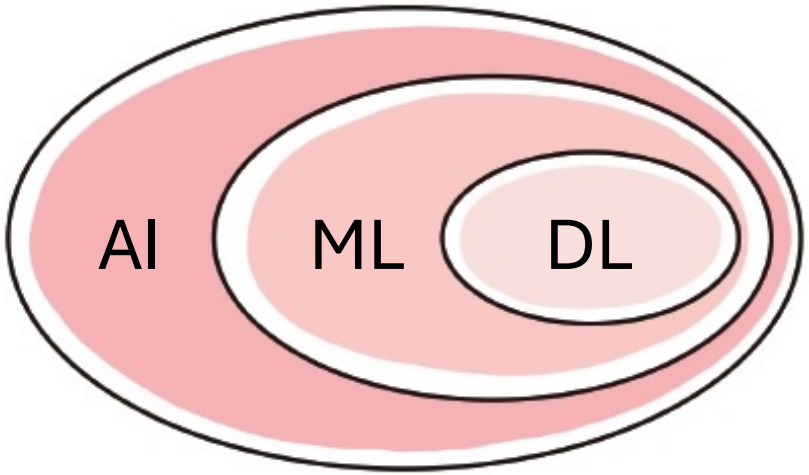
- 1997: IBM's Deep Blue
  - Defeated the world chess champion
  - A landmark moment for AI



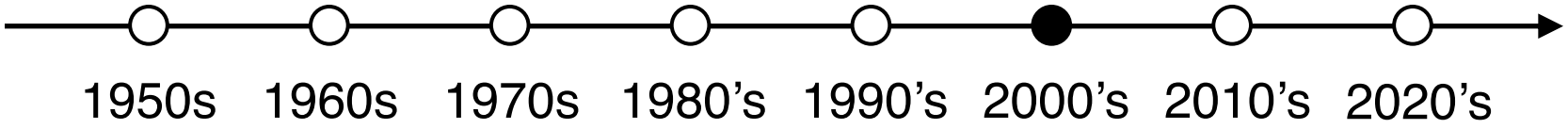
History of AI



- 2000s: Foundations of Deep Learning Established
  - AI, Aritifical Intelligence
  - ML, Machine Learning
  - DL, Deep Learning
    - Deep neural network + Machine learning

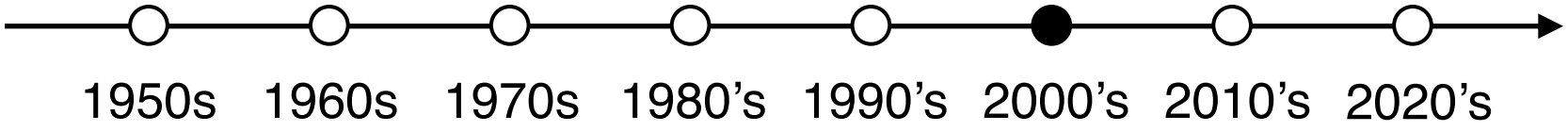


History of AI



- 2000s: Foundations of Deep Learning – Big Data
  - Internet: Widespread adoption since the mid-1990s
  - Smartphones: Apple iPhone launched in 2007
  - Social Media: Facebook (2004), YouTube (2005), Twitter (2006)

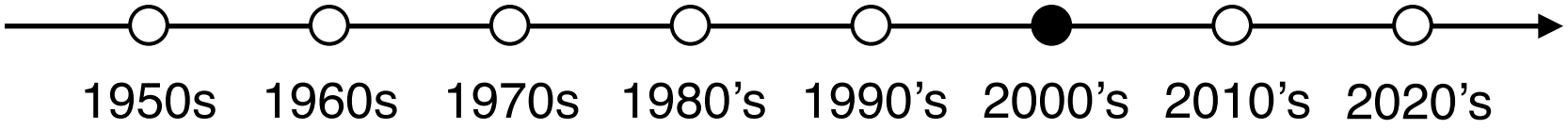
History of AI



- 2000s: Foundations of Deep Learning – GPU
  - 1999: NVIDIA introduced the GPU
  - 2006: NVIDIA introduced CUDA
  - From graphics processing to general-purpose computing

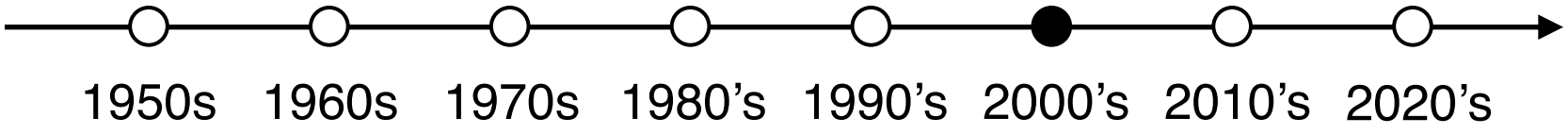


History of AI



- 2000s: Foundations of Deep Learning – Neural network
  - 1958: Perceptron
  - 1969: Limitations of single-layer perceptrons highlighted
  - 1986: Backpropagation for training multi-layer neural networks
  - 2006: Effective training of deep neural networks

History of AI

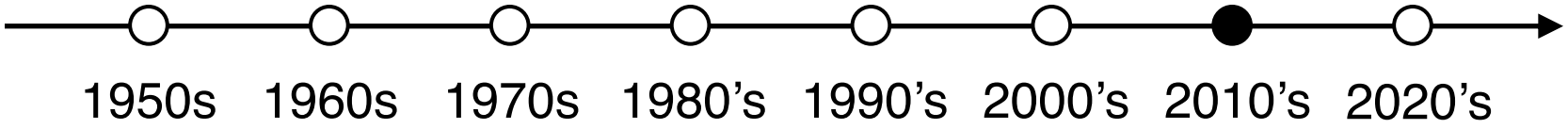


- 2000s: Foundations of Deep Learning – Neural network
  - Known as the "Godfather of Deep Learning"
  - Continued neural network research despite widespread skepticism
  - 2018: Turing Award
  - 2024: Nobel Prize in Physics

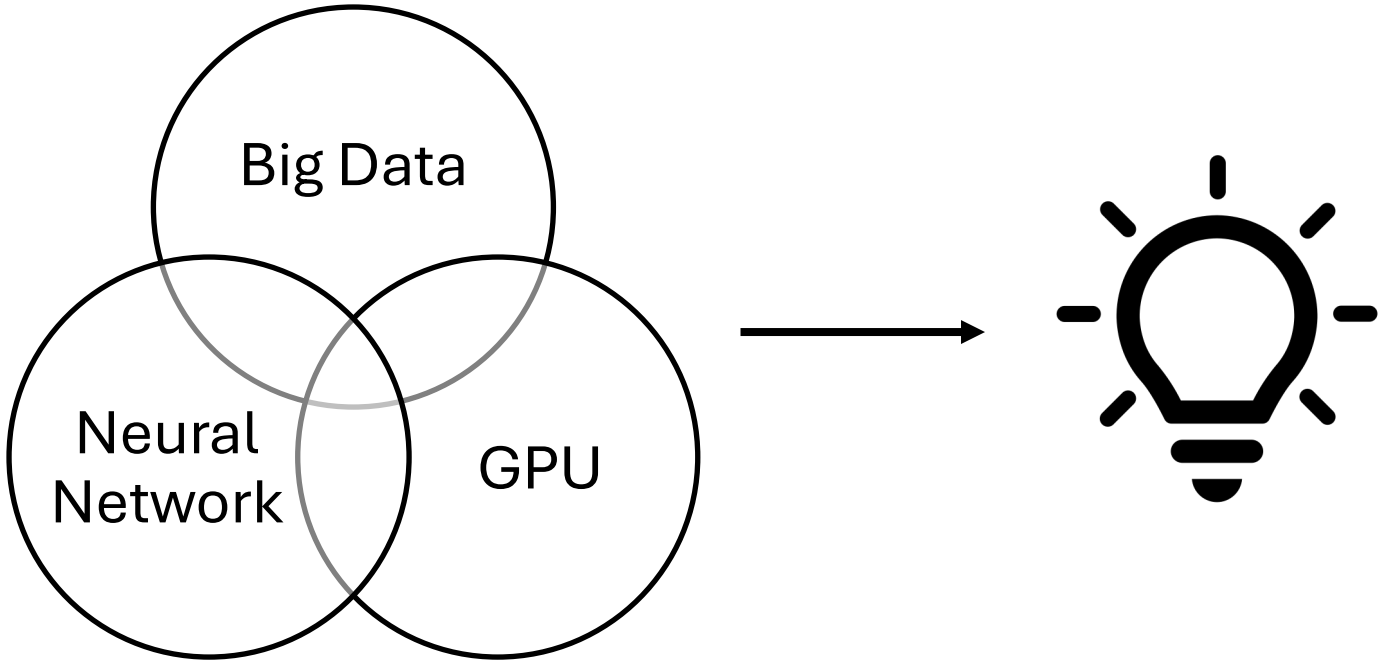
Geoffrey Hinton (1947–)



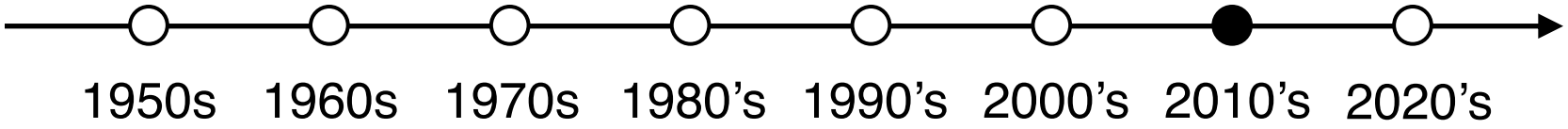
History of AI



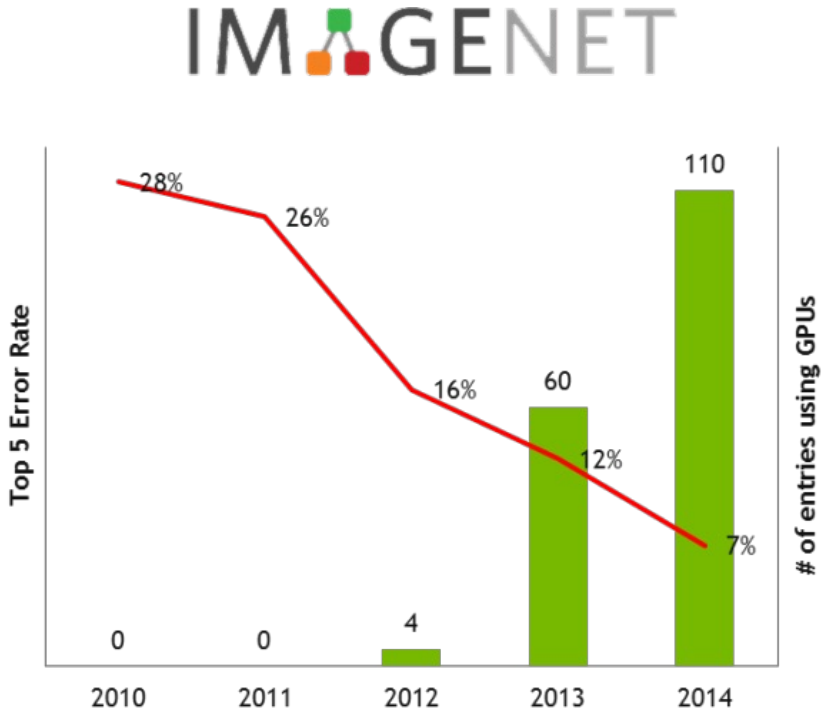
- 2010s: Deep Learning Revolution



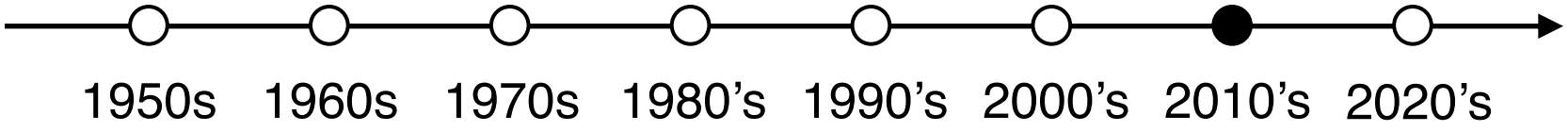
History of AI



- 2012: ImageNet Challenge
  - 1,000 categories
  - ~1.2 million images
  - Image classification: e.g., cat vs. dog
  - Deep learning dominates
  - Deep learning revolution begins



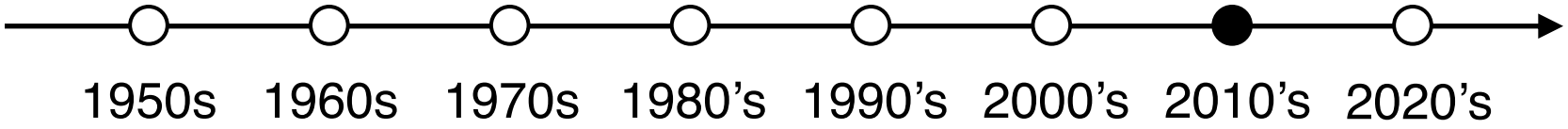
History of AI



- 2016: Google's AlphaGo
  - Defeated a world champion in Go
  - Mastered a highly complex game
  - Brought AI into the public spotlight



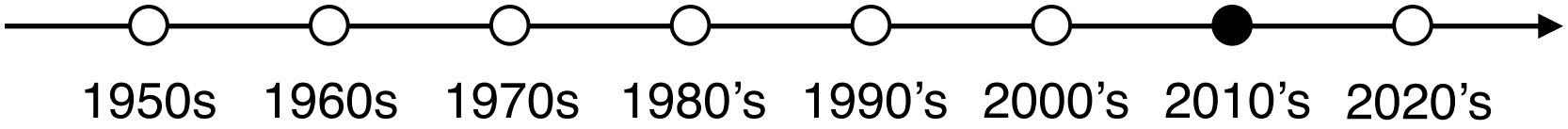
History of AI



- 2016: NVIDIA DGX-1
  - AI supercomputer
  - 8 GPUs
  - First DGX-1 delivered

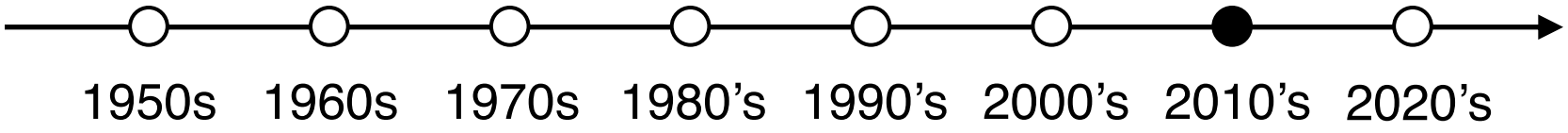


History of AI

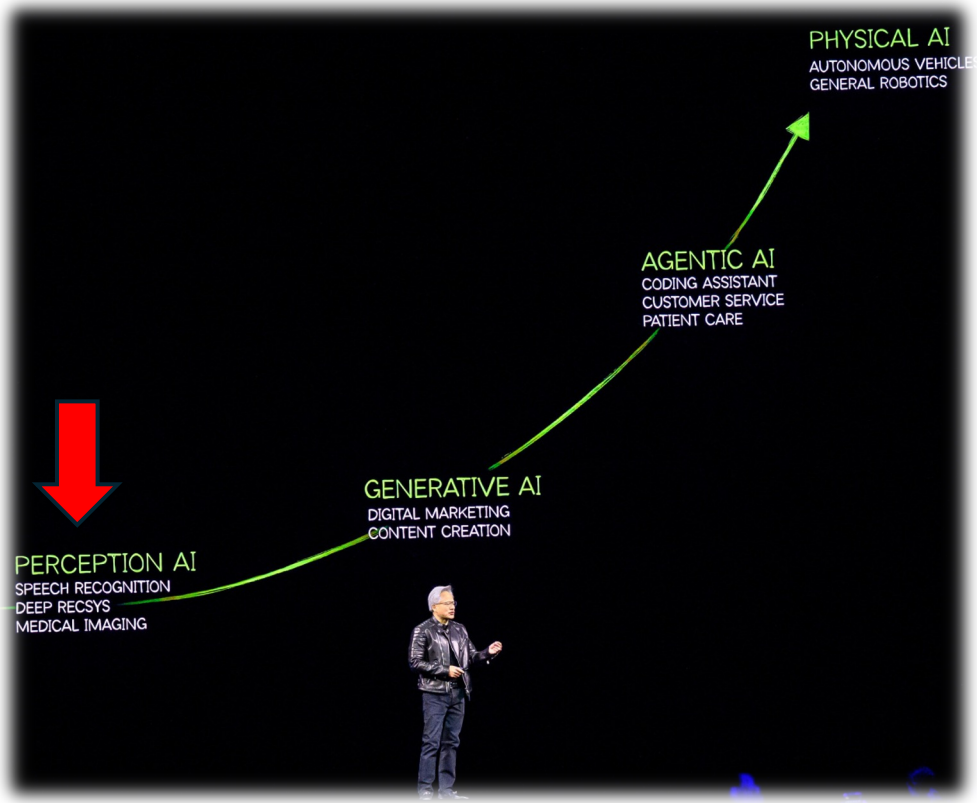


- 2017: Google's Transformer
  - Paper titled "Attention Is All You Need" published
  - Transformer architecture introduced
  - Foundation of modern AI models
    - Example: ChatGPT (Generative Pre-trained Transformer)

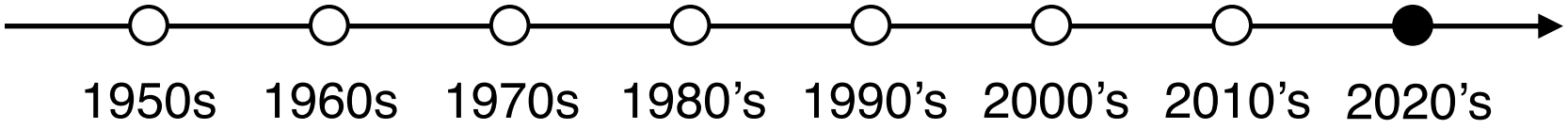
History of AI



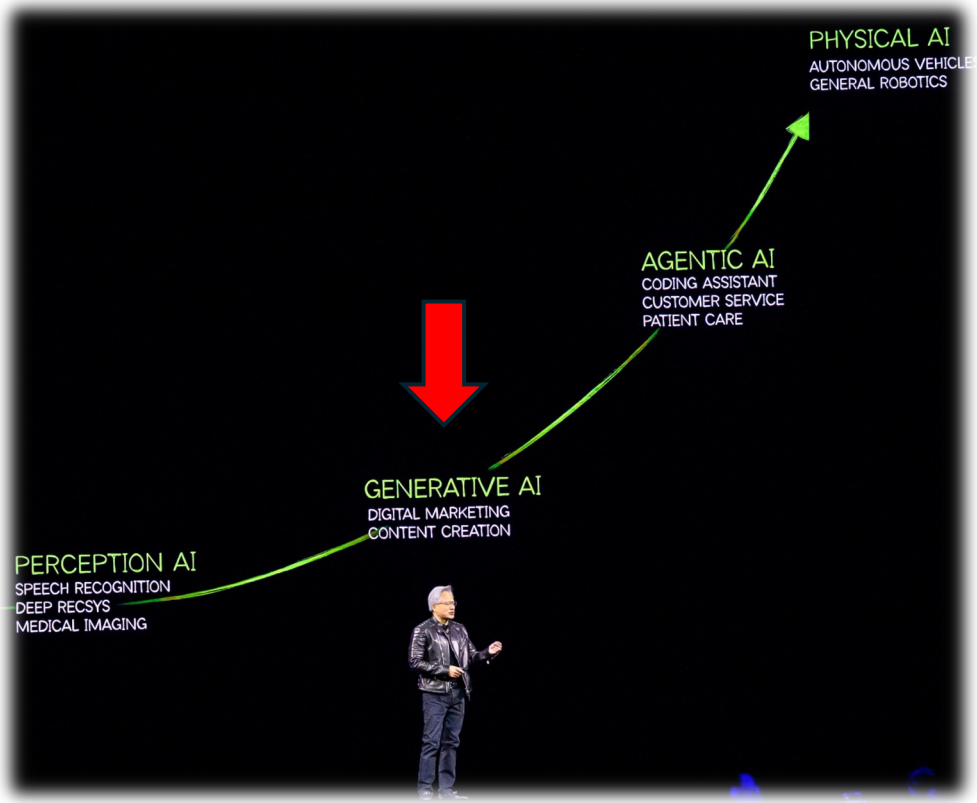
- 2010s: Perception AI
  - AI that perceives the world
  - Image classification
  - Object detection
  - Speech recognition



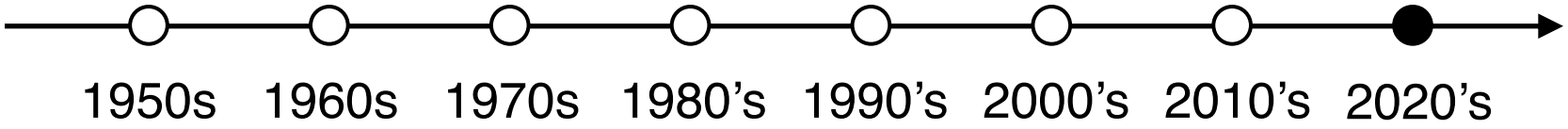
History of AI



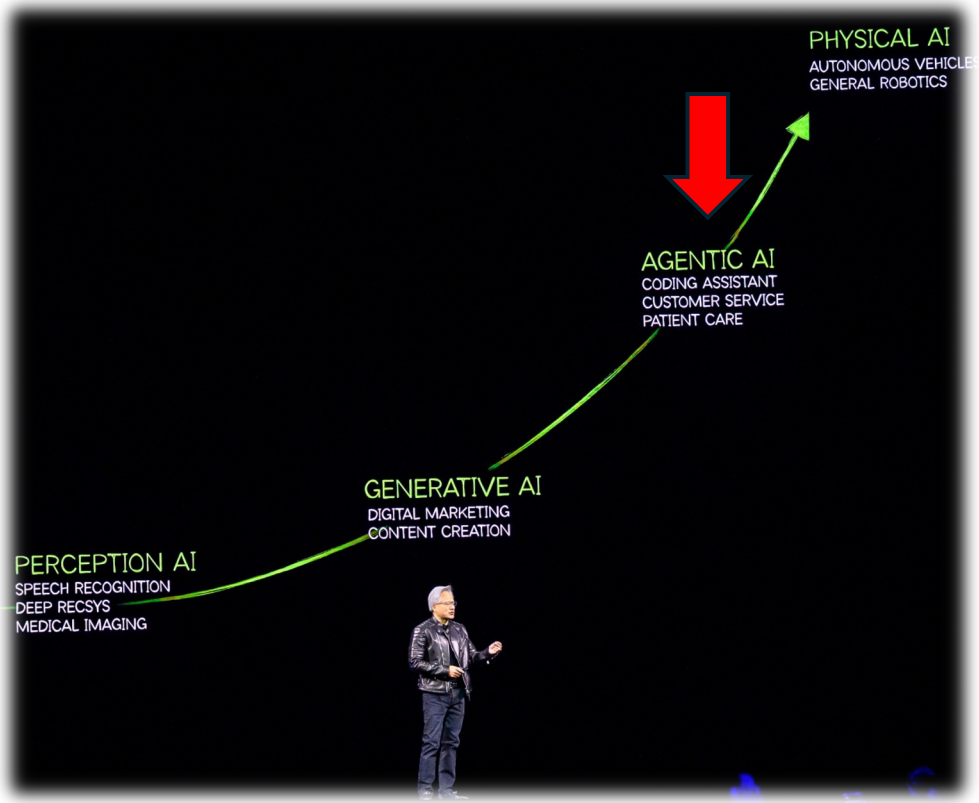
- 2020 – : Generative AI
  - AI that creates new content
  - Language, voice, and image
  - Example: ChatGPT



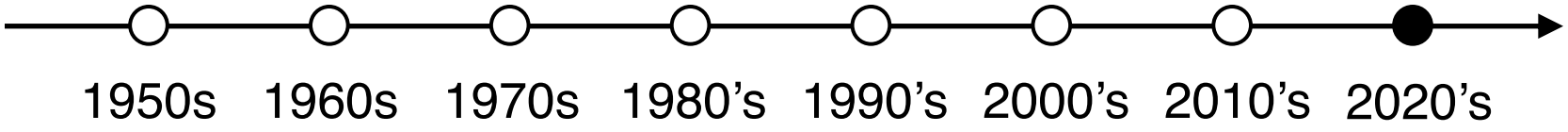
History of AI



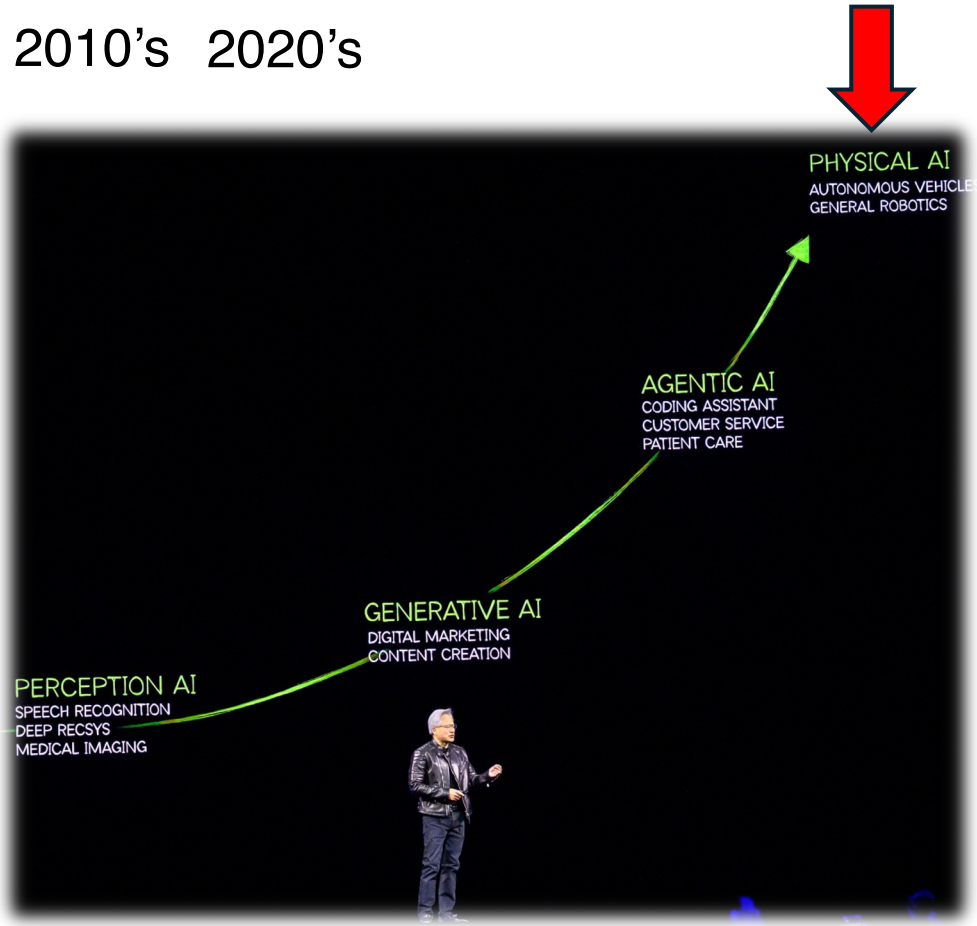
- 2023 – : Agentic AI
  - AI that performs tasks
  - Reasoning, Planning, Tool use
  - OpenAI Codex
  - Anthropic Claude Code



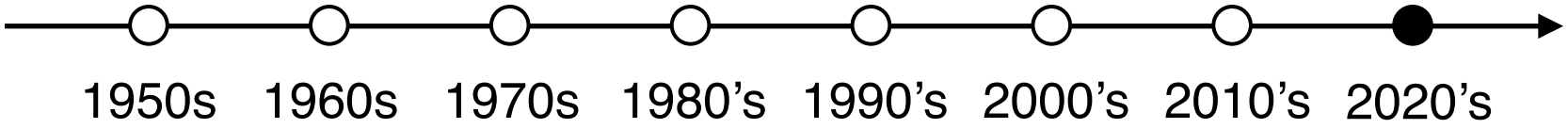
History of AI



- 2025 – : Physical AI
  - AI that acts in the physical world
  - Autonomous driving
    - Tesla FSD
  - Humanoid robots
    - Tesla Optimus



History of AI



- 2020s: AI Singularity?



"We're in the singularity."

"We'll hit AGI next year in 2026."

"By 2030, AI will exceed the intelligence of all humans combined."

"I think we've achieve AGI."

